Net change

1 In a closed habitat, the population of kangaroos $P(t)$ is known to increase according to the function:

$$P'(t) = \frac{t}{2} + 9$$

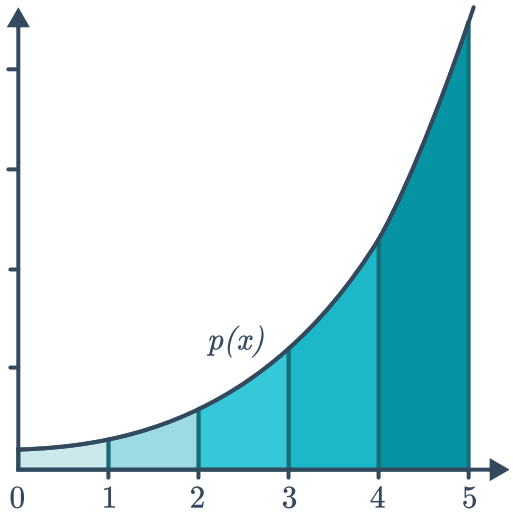
Where t is measured in months since counting began.

a Calculate the total change in the population of kangaroos in the first 4 months since counting began.

b Find the number of months it will take from when counting began for the population of kangaroos to increase by 88.

2 The graph shows the population growth rate of fish which is given by $p(x)$.
The total population growth of fish over n years is given by $\int_0^n p(x)\,dx$ as shown in the table below:

Year	Total growth
1	1000
2	4000
3	13 000
4	40 000
5	121 000



a Determine whether the following represent the total population growth in year 2.

i $\int_2^5 p(x)\,dx$

ii $\int_1^2 p(x)\,dx$

iii $\int_0^3 p(x)\,dx - \int_0^2 p(x)\,dx$

iv $\int_0^2 p(x)\,dx - \int_0^1 p(x)\,dx$

b Calculate the total population growth in year 2.

c Determine whether the following represent the total population growth from year 2 to 4.

i $\int_2^4 p(x)\,dx$

ii $\int_0^4 p(x)\,dx - \int_0^2 p(x)\,dx$

iii $\int_0^2 p(x)\,dx - \int_0^4 p(x)\,dx$

iv $\int_4^2 p(x)\,dx$

d Calculate the total population growth from year 2 to 4.

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3 An object is cooling and its rate of change of temperature, after t minutes, is given by:

$$T' = -10e^{-\frac{t}{5}}$$

a Find the instantaneous rate of change of the temperature after 8 minutes, to two decimal places.

b Find the total change of temperature after 9 minutes, to two decimal places.

c Hence, find the average change in temperature over the first 9 minutes, to two decimal places.

4 The average rate of change of function $f(x)$ over the domain $a \leq x \leq b$ is given by:

$$\frac{\int_a^b f'(x)\, dx}{b-a}$$

Calculate the average rate of change of the function $f(x)$ over the domain $\frac{\pi}{6} \leq x \leq \frac{\pi}{3}$ if:

a $f'(x) = 12\sin 6x$

b $f'(x) = 15\cos 3x$

5 Fluid leaks from a storage facility and its rate of change after t days is given by:

$$F'(t) = 3000e^{-0.5t}$$

a Find the amount of fluid that will leak in the first 7 days, correct to two decimal places.

b Find the amount of fluid that will leak in the next 7 days, correct to two decimal places.

6 Water flows into, then out of, a container at a rate $\frac{dV}{dt}$ litres per minute given by:

$$\frac{dV}{dt} = t(10-t)$$

Where the number of minutes, $t \geq 0$.

a Sketch the graph of $\frac{dV}{dt}$.

b Hence, find the maximum flow rate.

c Find an expression for the volume of water, V litres, in the container at time t minutes, assuming that the container is initially empty.

d Find the total time taken for the container to fill and then empty.

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7 The rate of emission $\frac{dE}{dt}$ of CFCs in Australia, measured in tonnes per year, was given by:

$$\frac{dE}{dt} = 100 + \left(\frac{60}{1+t}\right)^2$$

Where t is the time in years after 1 September 1988.

a Find the rate of emission on 1 September 1988.

b Find the rate of emission on 1 September 1997.

c Find the value that $\frac{dE}{dt}$ approaches as the years pass by.

d Find the total amount of CFCs emitted in Australia during the years 1988 to 1997.

8 Wheat is poured from a silo into a truck at a rate of $\frac{dM}{dt}$, measured in kilograms per second, given by:

$$\frac{dM}{dt} = 81t - t^3$$

Where t is the time in seconds after the wheat begins to flow.

a Find an expression for the mass M kg of wheat in the truck after t seconds, if initially there was 1 tonne of wheat in the truck.

b Find the total mass of wheat in the truck after 8 seconds.

c Find the largest value of t for which the expression for $\frac{dM}{dt}$ is physically possible.

9 The rate of flow of water into a supply tank is given by:

$$V' = 2000 - 30t^2 + 5t^3$$

For $0 \leq t \leq 7$, where V is the amount of water (in litres) in the tank t hours after midnight.

a Find the initial flow rate.

b Complete the table of values.

t	-1	0	1	4	6
V''					

c Hence, state the time t when the flow rate is a maximum, and the maximum flow rate at this time.

d Sketch the graph of V'.

e Find the area bound by the graph of V' and the t-axis between $t = 0$ and $t = 3$.

f Describe what the area found in part (e) represents.

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10 An ice cube with a side length of $25\ \text{cm}$ is removed from the freezer and starts to melt at a rate of $25\ \text{cm}^3/\text{min}$. Let V be its volume t minutes after it is removed from the freezer.

a State the equation for the rate of change of volume.

b State the equation for the volume, V, of the cube as a function of time t.

c Find the value of t when the ice cube has melted completely.

11 The mass, m grams, of a raindrop falling for t seconds is increasing at a rate given by:

$$\frac{dm}{dt} = \frac{1}{120}\left(\sqrt{t} + \frac{t^2}{12}\right)\ \text{g/s}$$

a Given that the initial mass of the raindrop is zero, find the mass m of the raindrop after t seconds.

b Find the exact mass of the raindrop after 16 seconds have passed.

c Another raindrop starts as a gas particle with a mass of $0.004\ \text{g}$. How much heavier will it be after 16 seconds than a raindrop that is initially weightless?

12 To determine the number of staff required to serve customers at a new cafe, a consultancy company monitored the number of customers over a ten hour period.
The rate of change of the number of customers seen during t hours of service was seen to be modelled by:

$$\frac{dC}{dt} = 50\pi\cos\left(\frac{\pi(t-3)}{20}\right)$$

a Approximately how many customers visited the cafe in the first hour?

b Approximately how many customers visited the cafe in the last two hours?

c Calculate the average number of customers per hour over the entire ten hour day.

Cost, revenue and profit applications

13 The marginal cost for the production of the xth item is modelled by:

$$C'(x) = 8x + 909\ \text{dollars per item}$$

a Find the net change in cost for producing between 13 and 19 items.

b Find the average change in cost for producing between 13 and 19 items.

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14 The marginal profit from the sale of the xth item is given by:

$$P'(x) = 0.0012x^2 + 1.2x - 4.1$$

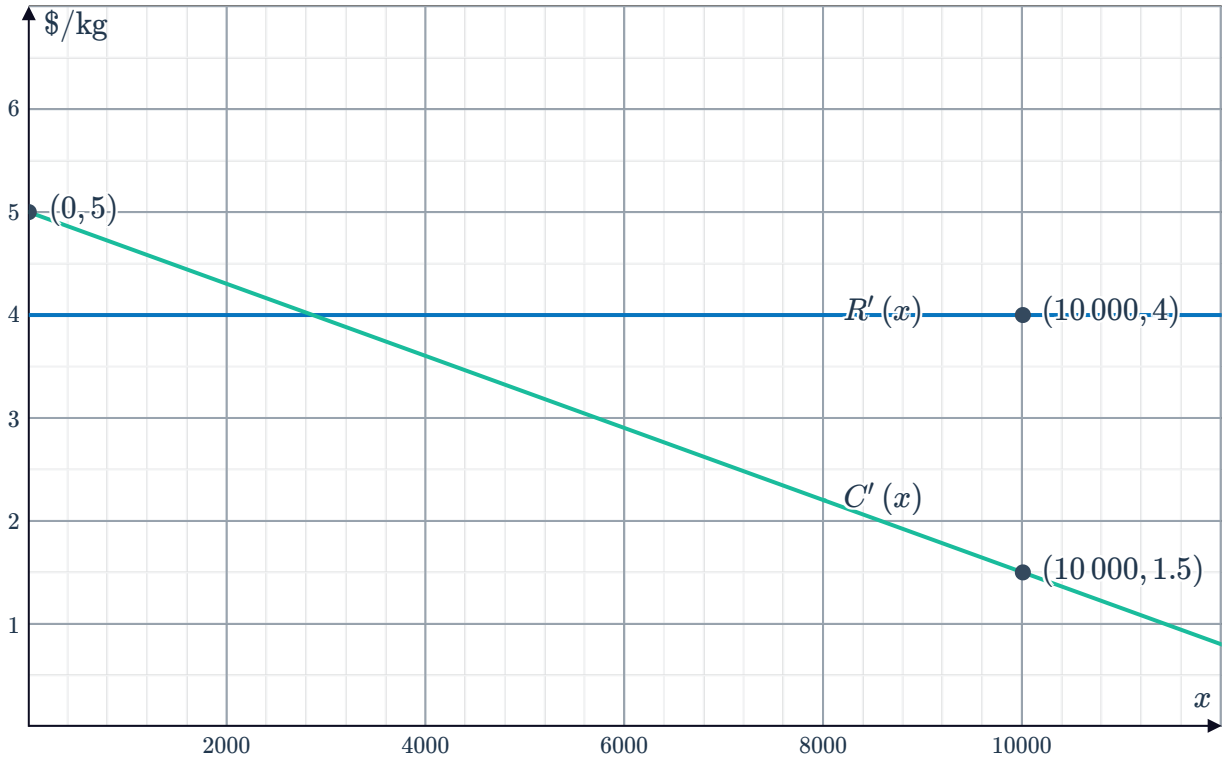
Where $P(x)$ is the profit from selling x items.

a Given that the company incurs a loss of $\$60$ if no items are sold, find an expression for P in terms of x.

b Hence, determine the profit from selling 70 items.

c Find the net change in profit if the number of items sold changes from 70 to 140 items.

15 The marginal cost, $C'(x)$, and marginal revenue, $R'(x)$, of a company producing x kg of coffee is plotted below:



a Find the area under each of the following functions over the interval $[0, 10\,000]$:

i $R'(x)$ **ii** $C'(x)$

b Hence, state the total revenue of producing the first $10\,000$ kg of coffee.

c Hence, state the total cost of producing the first $10\,000$ kg of coffee, given initial set-up costs of $\$2000$.

d The profit for selling x kg of coffee is given by $P(x) = R(x) - C(x)$. Determine the total profit of the first $10\,000$ kg of coffee.

e Market changes result in the marginal cost of coffee increasing by 20% while the marginal revenue decreases by 10%. Determine the total profit of the first $10\,000$ kg of coffee after the market change.

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16 $C(x)$ is the cost of producing x items of a certain product. The marginal cost of producing the xth item is given by:

$$C'(x) = 0.4x^3 + 4x$$

a Determine the cost function $C(x)$, in terms of x, given the fixed cost of production is $\$2000$.

b Hence, determine the cost of producing 50 items.

17 $C(x)$ is the cost of producing x units of a certain product. The marginal cost of producing the xth unit is given by:

$$C'(x) = 9x^2 - 70x + 500$$

a Determine the cost function $C(x)$ in terms of x.

b Hence, determine the extra cost incurred by producing 60 units rather than 20 units.

18 $C(x)$ is the cost of producing x units of a certain product. The marginal cost of producing the xth unit is given by:

$$C'(x) = \frac{120}{\sqrt{x}}$$

a Determine the cost function $C(x)$ in terms of x.

b Hence, determine the extra cost incurred by producing 81 units rather than 25 units.

19 $R(x)$ is the revenue from producing and selling x items. The marginal revenue from producing and selling the xth item is modelled by:

$$R'(x) = 2.7x(x - 200)$$

a Determine the change in revenue from producing and selling the 41st item.

b Determine the net change in revenue from selling 50 items to selling 80 items.

c Determine the average change in revenue from selling 50 items to selling 80 items.

20 A factory manufactures tractors. The marginal cost, $C'(x)$, and the marginal revenue, $R'(x)$, are given by:

$$C'(x) = 3x + 1, R'(x) = 9x + 6x^2 + 6$$

a Given that zero production results in zero profit, determine the profit function $P(x)$.

b Hence, determine the total profit from production and sale of the first 30 tractors.

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21 $C(x)$ is the cost of producing x units of a certain product. The marginal cost of producing the xth unit is given by:

$$C'(x) = \frac{440}{(x+5)^2}$$

a Determine the extra cost incurred by producing 30 items rather than 20 items.

b Determine the extra cost incurred by producing 50 items rather than 30 items.

22 R is the total revenue, in thousands of dollars, from producing and selling a new product, t weeks after its launch. The marginal revenue from producing and selling the xth item is modelled by:

$$\frac{dR}{dt} = 401 + \frac{500}{(t+1)^3}$$

a Given that the initial revenue at the time of launch was zero, state the revenue function.

b Find the average revenue earned over the first 5 weeks.

c Calculate the revenue earned in the 6th week.

23 For each of the following marginal profit functions, which give the profit, in dollars, of producing and selling x items of a product. Find:

i The extra profit associated with producing and selling the 101st item.

ii The net change in profit from producing and selling the first 100 items.

iii The average profit from producing and selling the first 100 items.

a $P'(x) = 30\pi \cos\left(\frac{\pi x}{600}\right)$

b $P'(x) = 20\pi \sin\left(\frac{\pi x}{600}\right)$

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